MIL-C-46666B
10 August 1973
SUPERSEDING
MIL-C-46666A
7 September 1962

MILITARY SPECIFICATION

CARTRIDGE, GRENADE: 7.62MM, NATO, RIFLE, M64

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a cartridge intended for propelling grenades from launchers attached to 7.62mm weapons.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-1168	- Lot Numbering of Ammunition

SPECIFICATIONS

Military

MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition Of

FSC 1330

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2.1 (Cont'd)

DRAWINGS

Munitions Command

C7553707	- Cartridge, Grenade: 7.62mm, NATO, Rifle, M64
C7643674	- Classification of Cartridge Case Defects
F10521851	- Packing and Marking; Cartridges, Grenade, 7.62mm, NATO, Rifle, M64; Cartons, Envelopes, Box, Ammunition, M2A1; Box, Wirebound

PUBLICATIONS

U. S. Army Munitions Command

AMCR 715-505 Volume 3	- Ammunition Ballistic Acceptance Test Methods Volume 3, Test Procedures for 7.62mm Cartridges

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-The cartridge shall comply with the requirements specified on Drawing C7553707, referenced specifications and the following:

3.2 Velocity.-The cartridge fired at the indicated temperatures shall propel grenades weighing 1.56 pounds plus or minus 0.01 pounds (1.56 ± .01) at velocities specified below. The velocity shall be measured at a point 5 feet 6 inches beyond the forward end of the grenade when fully positioned for launching.

Temperature Degrees Fahrenheit (°F)	Velocity Feet per Second (Ft/Sec) Average Velocity	Minimum Individual	Maximum Individual	Standard Deviation
65 to 75	160 ± 5			2
125 to 130		140	200	
-40 to -45		140	200	

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3.3 Function and casualty.-The cartridge shall function without casualty at temperatures of minus 40 degrees to minus 45 degrees F, at ambient temperature and at plus 125 degrees to plus 130 degrees F.

3.4 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.4.1 Metal defects.-The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.4.2 Foreign matter.-The cartridge shall be free of corrosion, stains discoloration, dirt, oil and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 200 cartridges. The sample shall be manufactured using the same materials, equipment, process and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result

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4.2.1.2 (Cont'd)

in sample disapproval. Determination as to acceptability of any initial production shall be based upon results of initial tests only, and no second tests shall be permitted on that initial production sample.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.-Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number in accordance with MIL-STD-1168. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The Acceptable Quality Level (AQL) for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent. All non-conforming cartridges shall be rejected.

4.3.2.1 Classification of defects.-The classification of defects shall be as specified in Table I.

TABLE I

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Cartridge				
1	Discolored, dirty, oily, smeared			X	
2	Corroded or stained, if etched		X		
	Case				
4	Round head		X		
5	Dent				X
6	Split case		X		
7	Perforated case	X			

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4.3.2.1 (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
8	Draw scratch				X
9	Scratch			X	
10	Beveled underside of head		X		
12	Scaly metal				X
13	No chamfer on head (rim)		X		
14	Fold			X	
15	Wrinkle			X	
17	Bulge			X	
18	Illegible or missing head stamp			X	
19	Defective head			X	
	Primer				
32	No primer	X			
33	Cocked primer	X			
34	Inverted primer	X			
35	Loose primer		X		
36	Nicked or dented primer			X	
37	No waterproofing material (primer pocket joint)			X	
38	Defective crimp			X	
	Mouth				
39	Missing wad	X			
40	Wad lacquer missing or inadequate		X		
41	Wad cannelure missing or incorrect		X		
42	Mouth crimp missing or incorrect			X	
43	Opening between wad and case wall		X		
	Gaging				
44	Cartridge profile failure (requiring more than 20 lbs. dead weight to insert in profile and alignment gage)		X		
45	Diameter of extractor groove			X	
46	Diameter of head			X	
47	Thickness of head, max.			X	
48	Thickness of head, min.		X		
49	Depth of primer		X		
50	Length to shoulder		X		
	Weighing				
51	Weight, min.	X			

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4.3.2.1 (Cont'd)

1/ Refer to MIL-STD-636 for visual standards for defects 1 through 38. While no photographs of grenade cartridges are included in this standard, the cartridge case defects illustrated in the 7.62mm cartridge section can be used. In the event of conflict between Table I of this specification and MIL-STD-636 as to defect classification, the classification specified in Table I shall apply.

4.3.3 Tests.-The test listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.-The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges selected shall be thoroughly mixed before being divided into the samples for the various tests.

TABLE II

Test	Temperature F 65° to 75°	-40° to -45°	125° to 130°	Requirement Paragraph
Velocity	25 1/	25 2/	25 2/	3.2
Function and Casualty 3/	--	--	--	3.3

1/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the initial test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot. Testing of a second sample shall not be permitted if the individual velocity of any cartridge in the initial sample is less than 140 or more than 200 feet per second.

2/ The variables sampling plan of MIL-STD-414, example C-4 for an AQL of 0.25 percent for the lower limit and an AQL of .65 percent for the upper limit shall be used for each sample of 25 cartridges. Failure of any cartridge to comply with the requirements of 3.2 or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot. No retest permitted.

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4.3.3.1 (Cont'd)

3/ The lot shall be rejected when function and casualty defects noted in the firing of the velocity test exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the initial sample exceeds the acceptance number for the initial sample of Table III but is equal to or less than the acceptance number for the cumulative sample (Table III), a second sample consisting of double the quantity specified for the velocity test, shall be fired in the weapon specified therfor. If the total number of defects in the combined first and second samples exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If in testing the second sample, defects other than those for which the second sample is being tested, should occur to the extent that they exceed the acceptance numbers for the cumulative sample, the lot shall be rejected.

4.3.3.2 Firing defects.-Firing defects and acceptance numbers shall be as specified in Table III.

TABLE III

Defects	Acceptance Number: First Sample	Acceptance Number: Cumulative (First & Second Sample)
1. Misfire	0	1
2. Primer leak		
a. Perforation in firing pin indent in primer cup	1	5
b. Escape of gas through primer cup other than 2a.	0	2
c. Escape of gas around primer cup more than 50% of periphery	1	2
d. Blown primer or primer falls out of pocket on retraction of bolt	0	1
e. Primer remains in pocket but is physically loose	0	2
3. Case casualties 1/		
a. Longitudinal split 2/		
(1) Body (J)	0	2
(2) Body (K)	0	1
(3) To head (L)	0	1
(4) Through head (M)	0	1
b. Circumferential rupture 2/		
(1) Partial, body (J)	0	2
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1
(4) Complete	0	1
4. Failure to extract	0	1

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4.3.3.2 (Cont'd)

1/ No penalty shall be charged against a fired case with mouth partially or completely burned. If, however, such fired cartridge case shows the occurrence of some other firing casualty, exclusive of failure of rosebud crimp to open, such other casualty shall be penalized in accordance with Table III.

2/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

4.3.4 Packaging, packing and marking inspection.-During or immediately prior to the packaging operation, 100 percent examination of the cartridge shall be performed to ascertain that the cartridge type conforms to the drawing. The presence of a dummy, blank or any bullet type cartridge shall be classed as a critical defect. Occurrence of any incorrect type other than those listed shall be classed as a major defect. All non-conforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

4.3.5 Inspection equipment.-Except as otherwise provided by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.4 Test methods and procedures.

4.4.1 Velocity.-The tests shall be conducted in accordance with AMCR 715-505, Volume 3, except that weapon shall be allowed to cool to ambient temperature after a maximum of twenty-five cartridges have been fired and a proof slug shall be used in lieu of a practice grenade.

4.4.1.1 Low temperature test.-The test shall be conducted in accordance with AMCR 715-505, Volume 3 and the following: The ammunition shall be placed in a controlled temperature box in such a manner that all cartridges are subjected to a uniform temperature, cooled to a temperature of minus 40 to minus 45 degrees F., and held thereat for a minimum of four hours immediately before firing. The controlled temperature box shall be of sufficient capacity to permit cooling of the ammunition to the desired temperature, and to allow free circulation of air. The weapon in which the ammunition is fired shall be at ambient temperature.

4.4.1.2 High temperature test.-The test shall be conducted in accordance with AMCR 715-505, Volume 3 and 4.4.1.1, except that the temperature at which the ammunition is conditioned shall be plus 125 to plus 130 degrees F.

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4.4.2 Function and casualty.-Observation for firing defects listed in Table III shall be made during the firing of the velocity test. Retest shall be fired in accordance with the procedure prescribed for the velocity test in AMCR 715-505, Volume 3, except that all references pertaining to obtaining of velocity data shall be disregarded.

4.4.3 Defect penalty.-If, in the velocity test, the occurrence of a firing defect prevents the obtaining of a reliable result, an additional shot shall be fired. The velocity test shall not be penalized, but the acceptance sample shall be penalized for such defects in accordance with Table III.

5. PREPARATION FOR DELIVERY

5.1 Packing. - Level A - (Worldwide Shipment) The cartridge shall be packed in accordance with Drawing F10521851, or as required by contract.

5.2 Marking and labeling.-Packing boxes shall be marked and labeled in accordance with the applicable drawing cited in 5.1.

6. NOTES

6.1 Ordering data.-Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provisions for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.4 Provision for the submission of acceptance of inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.5 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

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6.2 Hazard notice.-The cartridge described herein and certain of its components are flammable and/or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

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Preparing Activity:

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Project No. 1330-0110

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